

Oleg Sidorenkov

Senior SWE / Tech Lead

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Summary

Senior Software Engineer, 4+ years of experience

- Backend, large-scale distributed systems, security, and cloud
- Go, PostgreSQL, Kafka, Linux, Kubernetes, Vault, Terraform

Tech Lead, KMS Infrastructure, 1+ years of experience

- E2E ownership and cross-team delivery of platform products
- Mentoring and hiring engineers (5+ hires, 50+ interviews)
- Security Champion; driving DevOps practices within the team
- Public speaker at IT conferences; tech writer; open-source contributor

Experience

VK Cloud Senior SWE / Tech Lead *2025.04-present*

KMS (PaaS), Senior SWE -> KMS (PaaS) Tech Lead
Architecture & Leadership:

- **Designed a multi-datacenter architecture for a cloud KMS based on Hashicorp Vault/OpenBao + designed API. 5 data-centers, 4 envs, both bare-metal and K8s**
- Delivered KMS and S3 teams integration in bucket encryption: gathered requirements, created ADR, built environments, and launched MVP
- **Delivered KMS and K8s teams integration in ETCD encryption:** gathered requirements, created multiple ADRs/MVPs, and launched into production
- Conducted over 20 interviews. **Hired 3 Middle/Senior Go devs for PaaS teams**
- Reduced bus factor through architecture documentation, duty handbooks, shared technical ownership, and mentoring engineers

Development:

- **Led the design and implementation of a Control-Plane that automates Vault cluster lifecycle and configuration:** initialisation, unsealing, provisioning of various entities (plugins, secrets, keys, users, policies, etc...)
- **Created a Terraform provider for managing lifecycle of secrets/keys in the cloud**
- Built a KMS Data-Plane exposed as VK Cloud KMS secrets/keys API
- **Forked and modified a K8s plugin for encrypting resources stored in ETCD**
- **Improved pods cpu usage across services by ~15% and memory usage by ~25%** via pprof research, tuned token types and lifetimes, removed redundant projects

Operations & Security:

- **Integrated ETCD encryption into Managed Kubernetes cluster** based on Vault and K8s plugins
- Created Terraform modules for easier secrets migration from other clouds
- **Owned delivery, security, and observability of KMS services:**
  - Built and hardened rootless distroless multistage Docker images
  - Implemented GitLab CI/CD pipelines: Building, Testing, Linting, Scanning vulnerabilities/CVEs, and Delivery to K8s clusters via HelmWave
  - Set up monitoring: metrics and alerting (VictoriaMetrics + AlertManager)
  - Delivered Helm charts for backend services
- **Fixed over 100 vulnerabilities across services (misconfigs, deps, secrets, RBAC)**
- Duty: Incident (on-call) and operations (asynchronous help in chat)

Ozon SWE *2023.11-2024.12*

Seller's Actions (FinTech), Middle SWE
Architecture & Leadership:

- Redesigned PostgreSQL sharding strategy. **Average number of live connections dropped from 50 to 30. Largest table row count reduced by 90%**
- Coordinated 2 teams (analytics and development) during Kafka events API migration. **The change resulted in reducing latency by 50% in p50, by 80% in p90, and by 90% in p99**
- Conducted over 30 technical interviews. **Hired 3 Go devs for our team**

Development:

- **Maintained Go-microservices product serving 5,000+ RPS**
- Built an API that allows sellers to communicate with the system via Excel
- Migrated PostgreSQL (re-implemented all queues, implemented migrator)
- Reduced architectural technical debt and removed legacy components:

Operations:

- Created GitLab CI pipelines such as ProtoLint static check
- Duty: Incident (on-call) and operations (asynchronous help in chat)

Yandex SRE / SWE *2022.07-2023.11*

NOC, Intern SWE -> Market, Junior SRE -> Market, Middle SRE
Development:

- Maintained a warehouse service, that monitors states of CUPs/NUCs/PDTs. Go, PostgreSQL, ReactTS
- Maintained firewall config service. Go, Python, React, MongoDB. Led the integration with Yandex.Calendar API

Operations:

- Set up CI/CD pipelines and monitoring
- Duty: Incident (on-call) and operations (asynchronous help in chat)

Skills

Development:

- **Go** (Backend, distributed systems, CLIs, automation, pprof)
- Rust, C, Shell/Bash, Python
- Databases (**PostgreSQL**, ClickHouse, Redis, MongoDB, ETCD, S3)
- Queues/Event Streaming (**Kafka**)

Operations:

- **Linux** (administration, containerisation, networking, ptrace, strace, perf)
- **Kubernetes** (Helm, CRD, CRI/CNI/CSI, Docker images)
- Networking (TCP/IP, HTTP, gRPC, NetBox, eBPF, DPDK)
- CI/CD (GitHub Actions, GitLab CI/CD, Jenkins, ArgoCD)
- IaC (Terraform, Ansible)
- Cloud (OpenStack, VK Cloud, Yandex Cloud)
- Observability:
  - Metrics (Prometheus, VictoriaMetrics, Grafana)
  - Logging (Zap, OpenTelemetry, ELKB)
  - Tracing (OpenTelemetry, Jaeger)
  - Alerting (AlertManager, Sentry)

System Design & Architecture:

- API (OpenAPI/Swagger, protobuf)
- High Availability & High Load (CAP, caching, L4/L7 load balancing, profiling)
- Fault Tolerance (DRP, graceful shutdown, graceful degradation, backoff)
- Microservices & Async Intercommunication

Security:

- Keys and secrets (**HashiCorp Vault**, OpenStack Barbican)
- K8s (RBAC, Kyverno, image hardening, non-root, distroless)
- Scanners (Trivy, Linters)

Testing:

- Unit/Fuzzing (mocks)
- Integration/Acceptance/E2E (TestContainers, Terraform)

Education & Certifications

Education:

- PhD student (part-time), Economics, 2024-present
- **Master's degree, Computer Science, HSE FCS, 2024-2026, GPA 8.1/10**
- Bachelor's degree, Computer Science, HSE FCS, 2020-2024, GPA 8.65/10
- External student for 1 semester at THWS, 2023

Certifications:

- Tinkoff SRE

Talks & Publications

Speaker:

- "Contextual Talk", Evrone Go Meetup
- "Obscure STL Containers and Where to Apply Them", E-CODE
- "Container Internals or how 4 Syscalls Changed the Operations", Stachka

Writer:

- "Container Internals or how 4 Syscalls Changed the Operations", Habr

Open-source & Pet-projects

Terraform Provider VKCS

Terraform provider for managing VK Cloud entities. I've implemented KMS resources and data sources as well as provided modules (e. g. migration from other clouds)
Stack: Go, HashiCorp Vault, HashiCorp Terraform

Vault Kubernetes KMS

K8s Control Plane component that encrypts Secrets and ConfigMaps via Vault. Thus, entities stored in ETCD are secured. I've implemented userpass auth integration
Stack: Go, Kubernetes, HashiCorp Vault

DevSecMLOpsTemplate

Template for secure deployment of ML services based on CRDs/Terraform entities for integrating with trusted model registry and Kyverno policies over S3 and Vault
Stack: Go, K8s, CRD, Kyverno, Helm, GitLab CI/CD, AWS S3, ArgoCD, Terraform, Vault

Go-libs

Declarative type conversion, LRU-cache
Stack: Go

Mascoty

Cross-platform application for creating 2D-avatars with real-time facial expressions
Stack: Rust, TypeScript, Tauri, Torch

Teaching

HSE Instructor *2025.01-present*

Operating Systems course instructor. I conduct classes about UNIX/Linux internals

HSE & School №444 Teaching Assistant *2021.09-2022.05*

Algebra and Calculus Mathematical Practice in Matlab teaching assistant
Data Structures and Algorithms in Python teacher

Languages

🇷🇺Russian, native

🇬🇧English, C1, Academic IELTS - 7.5/9